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Candidate Number

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HONG KONG DIPLOMA OF SECONDARY
EDUCATION EXAMINATION BYTOPIC

MATHEMATICS Extended Part

Module 2 (Algebra and Calculus)

Question-Answer Book

Time allowed : $2\frac{1}{2}$ hours

This paper must be answered in English

INSTRUCTIONS

- (1) After the announcement of the start of the examination, you should first write your Candidate Number in the space provided on Page 1 and stick barcode labels in the spaces provided on Pages 1.
- (2) This paper consists of TWO sections, A and B.
- (3) This paper carried 100 marks.
- (4) Attempt ALL questions in this paper. Write your answers in the spaces provided in this Question-Answer Book. Do not write in the margins. Answers written in the margins will not be marked.
- (5) Graph paper and supplementary answer sheets will be supplied on request. Write your Candidate Number, mark the question number box and stick a barcode label on each sheet, and fasten them with string INSIDE this book.
- (6) Unless otherwise specified, all working must be clearly shown.
- (7) Unless otherwise specified, numerical answers must be exact.
- (8) The diagrams in this paper are not necessarily drawn to scale.
- (9) No extra time will be given to candidates for sticking on the barcode labels or filling in the question number boxes after the 'Time is up' announcement.

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FORMULAS FOR REFERENCE

$\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$	$\sin A + \sin B = 2 \sin \frac{A+B}{2} \cos \frac{A-B}{2}$
$\cos(A \pm B) = \cos A \cos B \mp \sin A \sin B$	$\sin A - \sin B = 2 \cos \frac{A+B}{2} \sin \frac{A-B}{2}$
$\tan(A \pm B) = \frac{\tan A \pm \tan B}{1 \mp \tan A \tan B}$	$\cos A + \cos B = 2 \cos \frac{A+B}{2} \cos \frac{A-B}{2}$
$2 \sin A \cos B = \sin(A + B) + \sin(A - B)$	$\cos A - \cos B = -2 \sin \frac{A+B}{2} \sin \frac{A-B}{2}$
$2 \cos A \cos B = \cos(A + B) + \cos(A - B)$	
$2 \sin A \sin B = \cos(A - B) - \cos(A + B)$	

SECTION A (74 marks)

1. In the expansion of $(1 + ax)^n$, the coefficient of x^{n-2} is 54 and the coefficient of x is -12 .

(a) Find a and n . (4 marks)

(b) Find the coefficient of x^3 . (2 marks)

Answers written in the margins will not be marked.

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2. In the expansion of $(2 + 4x)^n$, where n is a positive integer, let u_k be the coefficient of x^k .
It is given that $2u_1 + 8u_0 = u_2$.

(a) Find n . (3 marks)

(b) Hence, evaluate $0 \cdot u_0 + 1 \cdot u_1 + 2 \cdot u_2 + \cdots + n \cdot u_n$. (2 marks)

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3. Consider the expansion of $\left(1 + \frac{1}{x}\right)^2 \left(kx + \frac{1}{x}\right)^4$, where k is a constant. It is given that the constant term in the expansion is 0.

(a) Find the value(s) of k . (3 marks)

(b) Someone claims that for any positive integer n , the coefficient of x^{-n} in the expansion of $\left(1 + \frac{1}{x}\right)^2 \left(kx + \frac{1}{x}\right)^n$ must not be 1. Do you agree? Explain your answer.

(3 marks)

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write $(au + t)^n = \sum_{r=0}^n \alpha_r u^r$. It is given that $\alpha_1 = 12$, $\alpha_2 = 60$, and $\alpha_3 = 160$.

(a) Find a, t and n . (4 marks)

(b) Write $(au + bv)^n = \sum_{r=0}^n \beta_r v^r$, where b is an integer. Prove that $\beta_4 - \alpha_4$ is divisible by $b^2u + 2$. (3 marks)

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5. Consider the expansion of $(x^{-2} + x)^5(x^3 - x^{-2})^2$.

(a) Find the coefficient of x^{-4} in the expansion. (3 marks)

(b) Someone claims that the coefficient of x^2 in the expansion is 0. Do you agree? Explain your answer. (3 marks)

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6. Consider the expansion of $(1+kx)^n$, where n is an integer greater than 2, and k is a non-zero constant.

It is given that the sum of the coefficient of x and the coefficient of x^2 is 50. In addition, the ratio of the coefficient of x^2 to the coefficient of x^3 is 1 : 2.

(a) Find the values of n and k . (4 marks)

(b) Let $(1+kx)^n = P(x)$. If a constant term exists in the expansion of $\left(x^2 - \frac{1}{2x}\right)^6 P(x)$, find the value of this constant term. (4 marks)

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9. A sequence of functions $F_n(x)$ is defined by

$$F_n(x) = \sum_{k=1}^n \left[\binom{n}{k} \cdot \left(\frac{x}{1+2x} \right)^k \right].$$

- (a) (i) By considering the expansion of the binomial theorem, prove that

$$1 + F_n(x) = \left(\frac{1+3x}{1+2x} \right)^n.$$

(2 marks)

- (ii) Consider $F_n(x) = c_0 + c_1x + c_2x^2 + \cdots + c_nx^n$. Hence, find c_1 in terms of n .

(2 marks)

- (b) Consider another function $G_n(x) = (1+2x)^n \cdot [1 + F_n(x)]$.

- (i) Prove that $G_n(x) = (1+3x)^n$.

(1 mark)

- (ii) Let $G_n(x) = b_0 + b_1x + b_2x^2 + \cdots + b_nx^n$. Prove that for all $2 \leq r \leq n$,

$$\frac{b_r}{b_{r-1}} = \frac{3(n-r+1)}{r}.$$

(4 marks)

- (c) It is given that the ratio of three consecutive coefficients in the expansion of $G_n(x)$ is $b_{r-1} : b_r : b_{r+1} = 1 : 3 : 7$. Find the values of n and r .

(3 marks)

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SECTION B (26 marks)

10. It is given that matrix $M = \begin{pmatrix} 2 & -1 \\ 3 & -2 \end{pmatrix}$ and $I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$.

(a) Prove that $M^2 = I$. (2 marks)

(b) Consider the matrix series $X = \sum_{r=0}^5 \binom{5}{r} 2^{5-r} M^r$.

By using the binomial theorem and the result of (a), express the matrix X in the form $aI + bM$ (where a and b are constants), and find the exact value of the matrix X .

(5 marks)

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11. It is given that $A = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$ and $I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$.

(a) Prove by mathematical induction that for all positive integers n ,

$$A^n = \frac{1}{2} \begin{pmatrix} 3^n + (-1)^n & 3^n - (-1)^n \\ 3^n - (-1)^n & 3^n + (-1)^n \end{pmatrix}.$$

(5 marks)

(b) Consider a matrix B_n defined by

$$B_n = \sum_{r=0}^n \binom{n}{r} A^{2r}$$

where n is a positive integer.

(i) Prove that $A^2 - 2A - 3I = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$. (1 mark)

(ii) Without using comparison of coefficients, by expanding $(I + A^2)^n$ using the binomial theorem, prove that

$$B_n = \frac{1}{2} \begin{pmatrix} 10^n + 2^n & 10^n - 2^n \\ 10^n - 2^n & 10^n + 2^n \end{pmatrix}.$$

(4 marks)

(c) In the Cartesian plane, the three vertices of $\triangle OPQ$ are the origin $O(0,0)$, $P(1,0)$, and $Q(0,1)$. Two points P' and Q' satisfy the matrix equations:

$$\begin{pmatrix} x_{P'} \\ y_{P'} \end{pmatrix} = B_n \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} x_{Q'} \\ y_{Q'} \end{pmatrix} = B_n \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

where B_n is the matrix defined in (b)(ii).

(i) Using the result of (b)(ii), find the determinant of B_n , $\det(B_n)$, and express your answer as a single exponential expression. (2 marks)

(ii) It is given that the area of $\triangle OP'Q'$ can be calculated by $\text{Area} = \frac{1}{2}|x_{P'}y_{Q'} - x_{Q'}y_{P'}|$. Prove that the area of $\triangle OP'Q'$ is 20^n times the area of $\triangle OPQ$. (2 marks)
